

ADITYA GAWANDE

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PROFESSIONAL SUMMARY

Generative AI & Machine Learning Engineer with 1.9+ years of production experience designing autonomous agent architectures, multi-modal RAG systems, and enterprise process automation. Hands-on expertise in fine-tuning open-source LLMs (LoRA/QLoRA), implementing Model Context Protocol (MCP) toolchains, and orchestrating LangChain & LangGraph workflows. Proficient in high-throughput inference serving with FastAPI and Docker/Kubernetes on Azure and GCP.

TECHNICAL SKILLS

Generative AI & LLMs:	LangChain, LangGraph, CrewAI, MCP (Model Context Protocol), RAG, Prompt Engineering, Function Calling, PEFT/LoRA Fine-Tuning, Ragas, NeMo Guardrails
AI / ML & Deep Learning:	PyTorch, TensorFlow, Hugging Face, Transformers, Scikit-Learn, CNNs, RNNs, Statistical Modeling, NLP (SpaCy, NLTK)
Vector DBs & Retrieval:	ChromaDB, FAISS, Pinecone, PostgreSQL (pgvector), Hybrid Search, Semantic Re-ranking
Backend & Cloud:	Python, FastAPI, Flask, Node.js, WebSockets, REST APIs, GCP Cloud Run, Azure AI Foundry, Docker, Kubernetes, CI/CD

PROFESSIONAL EXPERIENCE

LTIMindtree – Technical Consultant (AI & GenAI Solutions)

Dec 2024 – Present | Pune, India

- Architected an enterprise Autonomous Email Assistant using **Microsoft Copilot Studio**, **Power Automate**, and **Azure Functions**, automating contextual replies across PDFs, SharePoint, and URLs; cut triaging time by **65%**.
- Engineered a low-latency real-time RAG engine with **LangChain**, **GPT-4**, and **ChromaDB** over **WebSockets**, hosted via **GCP Cloud Run** with sub-800ms semantic search response.
- Developed a real-time SAP IDOC error diagnosis and resolution pipeline with **Human-in-the-Loop (HITL)** approval gates, dynamic retry automation, and automated alerts via **Azure Communication Services**.
- Integrated semantic caching and prompt optimization pipelines, reducing duplicate LLM API invocations and trimming inference expenses by **30%**.

KEY PROJECTS

Autonomous Multi-Agent Enterprise Research Analyst

LangGraph, MCP, ChromaDB, Claude/GPT-4, FastAPI

- Engineered an autonomous multi-agent hierarchy utilizing **LangGraph** featuring specialized Planner, Code-Execution, and Reviewer sub-agents with stateful checkpointing in PostgreSQL.
- Integrated custom **Model Context Protocol (MCP)** servers for secure code-sandboxing and dynamic vector queries, eliminating prompt leakage.
- Implemented automated factuality verification and synthetic benchmark evaluation with **Ragas**, achieving a **96%** hallucination resistance rate.

Domain-Specific LLM Fine-Tuning & Quantized Serving

PyTorch, Hugging Face, QLoRA, vLLM, Ollama, Docker

- Fine-tuned **Llama-3 (8B)** and **Mistral (7B)** on domain enterprise datasets using **QLoRA (4-bit quantization)**, boosting domain terminology accuracy by **28%** on evaluation splits.
- Containerized and served models using **vLLM** with continuous batching and PagedAttention on Kubernetes, achieving **3.2x higher throughput** over naive inference.

MCP-Based Natural Language to SQL Analytics Engine

FastMCP, Gemini API, PostgreSQL, Python

- Built a high-fidelity Natural Language-to-SQL translation engine utilizing **FastMCP Server** and **Gemini Function Calling** to dynamically query schemas and execute sanitized relational queries.
- Integrated AST-based SQL query validation to enforce read-only execution guardrails, reducing ad-hoc reporting turnaround time by **75%**.

Intelligent Healthcare Report Analysis & Recommendation

CrewAI, FAISS, PyMuPDF, LangChain

- Engineered an autonomous diagnostic report parser with **CrewAI** processing complex lab PDFs with **98% data extraction accuracy**.
- Synthesized patient biomarker values against clinical knowledge graphs to produce tailored dietary guidelines, trialed across **500+ patient records**.

EDUCATION

DY Patil College of Engineering, Akurdi – B.E. in Artificial Intelligence and Data Science CGPA: 8.45/10 | Jun 2024 | Pune, India